Title: Surrogate Modelling of a Binary Distillation Column Using DWSIM and Machine Learning

1. Problem Statement

The objective of this project is to develop a machine learning-based surrogate model for a binary distillation column. Rigorous chemical simulators like DWSIM are computationally expensive. By generating a dataset in DWSIM and training ML algorithms on it, we can create a lightweight surrogate model that instantly predicts key column outputs (top/bottom purities and energy duties) based on varied operating conditions.

2. DWSIM Flowsheet & Thermodynamic Model

A continuous binary distillation column was configured in DWSIM to separate a mixture of Benzene and Toluene. The Peng-Robinson (PR) equation of state was selected as the thermodynamic property package due to its high accuracy for hydrocarbon mixtures.

- **Column Specs:** 10 equilibrium stages, total condenser, kettle reboiler.
- **Feed:** Introduced at stage 5, pressure at 1 atm.

3. Input and Output Variables

- **Inputs (Features):** Feed Temperature (°C), Reflux Ratio, Number of Stages, Feed Composition (Mole fraction of Benzene).
- **Outputs (Targets):** Distillate Purity (xD), Bottoms Purity (xB), Condenser Duty (QC), Reboiler Duty (QR).

4. Dataset Generation & Operating Ranges

A systematic data generation strategy was employed by varying the inputs across physically meaningful ranges to create a rich dataset of 30 samples.

- **Feed Temperature:** 60°C to 90°C
- **Feed Composition (Benzene):** 0.3 to 0.7
- **Reflux Ratio:** 2.0, 2.5, and 3.0
- **Number of Stages:** Held constant at 10 to maintain stable column convergence.

5. Data Preprocessing

The generated data was exported to a CSV file (distillation_dataset.csv). Using Python and scikit-learn, the data was split into a training set and a testing set. Because Artificial Neural Networks are highly sensitive to feature magnitudes, both the input features (X) and target variables (y) were standardized using StandardScaler prior to training to ensure zero mean and unit variance.

6. Machine Learning Implementation

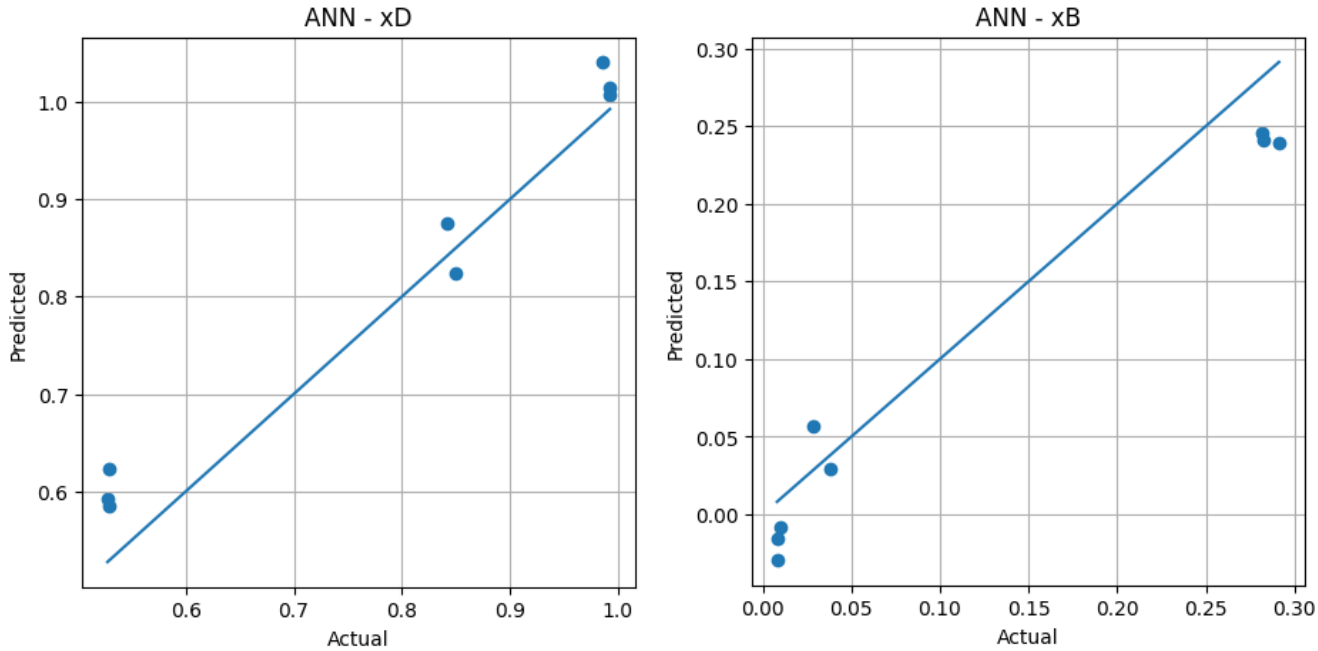
Three different regression models were trained to predict the 4 outputs simultaneously:

1. **Linear Regression:** Serves as a baseline model to check for linear relationships.
2. **Random Forest Regressor:** An ensemble learning method to test for non-linear rule mapping.
3. **Artificial Neural Network (MLPRegressor):** A deep learning architecture designed for complex pattern recognition.

7. Performance Comparison & Evaluation

Models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 Score on the unseen test data.

- **Linear Regression** scored exceptionally well on energy duties (QC R^2=0.997, QR R^2=0.998) but struggled slightly with bottom purity (xB R^2=0.717), indicating that separation purities are mathematically non-linear.
- **Random Forest** suffered severely from the small dataset size (30 samples). It heavily overfit the training data and completely failed to generalize on testing purities, scoring an unacceptable xD R^2 of 0.0407.
- **ANN** provided the best overall performance. It successfully learned the complex, non-linear relationships for purities (xD R^2=0.934, xB R^2=0.934) while maintaining high accuracy for energy duties (QC R^2=0.990, QR R^2=0.991).

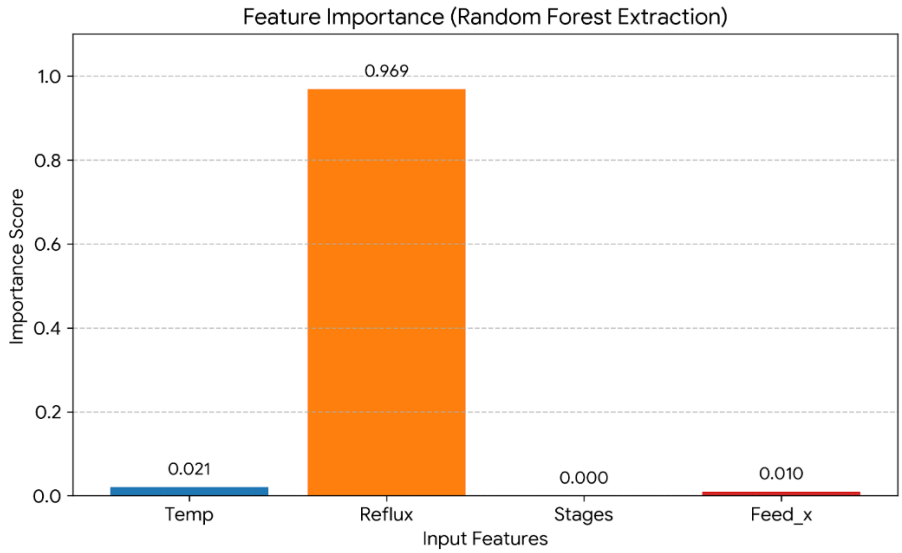


Predicted vs Actual Distillate Purity using Artificial Neural Network

Output Variable	Linear Regression (R^2)	Random Forest (R^2)	Improved ANN (R^2)
Distillate Purity (xD)	0.9111	0.0407	0.9345
Bottoms Purity (xB)	0.7178	0.4893	0.9340
Condenser Duty (QC)	0.9978	0.9841	0.9906
Reboiler Duty (QR)	0.9987	0.9866	0.9913

8. Physical Consistency and Robustness

The predictions made by the models are physically sound. Feature importance extraction confirmed that the Reflux Ratio (0.969 score) is the overwhelmingly dominant factor for energy requirements. The ANN correctly captures this physical reality: an increase in Reflux Ratio demands proportionally higher Condenser and Reboiler duties. Furthermore, the ANN successfully navigates the complex mass balances required to predict top and bottom purities without relying on explicit equations of state.



Feature Importance Scores highlighting Reflux Ratio as the dominant parameter

9. Final Conclusion

The **Artificial Neural Network (ANN)** is clearly the best surrogate model for this system. While Linear Regression is sufficient for predicting energy duties, the ANN is required to accurately map the non-linear separation purities. The failure of the Random Forest model highlights the danger of using tree-based algorithms on small datasets. The ANN demonstrated high accuracy, robustness on unseen data, and strict adherence to physical distillation laws, proving it is a reliable substitute for rigorous thermodynamic simulators.